GENERAL INSTRUCTIONS

1. The test is of 3 hours duration and the maximum marks is 300.

2. The question paper consists of 3 Parts (Part I: Physics, Part II: Chemistry, Part III: Mathematics). Each Part has two sections (Section 1 & Section 2).

3. Section 1 contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

4. Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted.

   The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the sign for positive values. However, for negative values, Θ sign should be bubbled. (Example: 6, 81, 1.50,⊕ 3.25, 0.08)

5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.

6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.

7. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

8. Do not fold or make any stray mark on the Answer Sheet (OMR).

Marking Scheme

1. Section – 1: +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

2. Section – 2: +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : …………………………………………………

Roll Number : …………………………………………………………………………

OMR Bar Code Number : ………………………………………………………………

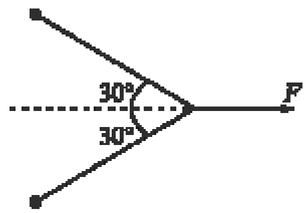
Candidate's Signature : ……………… Invigilator's Signature ……………………………..

PART - I : PHYSICS **100 MARKS**

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

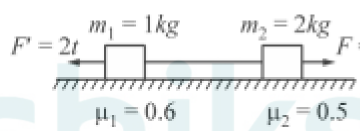
1. In figure, two identical particles each of mass m are tied together with an inextensible string. This is pulled at its centre with a constant force F. If the whole system lies on a smooth horizontal plane, then the acceleration of each particle towards each other is :



2.

(a) $\frac{\sqrt{3}}{2}\frac{F}{m}$ (b) $\frac{1}{2\sqrt{3}}\frac{F}{m}$ (c) $\frac{2}{\sqrt{3}}\frac{F}{m}$ (d) $\sqrt{3}\frac{F}{m}$

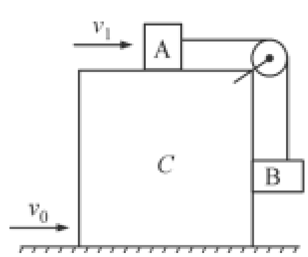
2. Two blocks A and B are separated by some distance and tied by a string as shown in the figure. The force of friction on both the blocks at t = 2s is:



(a) $4N(\rightarrow), 5N(\leftarrow)$ (b) $2N(\rightarrow), 5N(\leftarrow)$

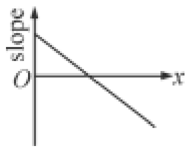
(c) $0N(\rightarrow), 10N(\leftarrow)$ (d) $1N(\leftarrow), 10N(\leftarrow)$

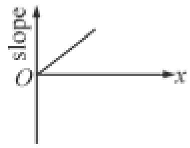
3. To a ground observer the block C is moving with v_0 and the block A with v_1. B is moving with v_2 relative to C as shown in the figure. Identify the correct statement.

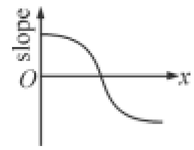


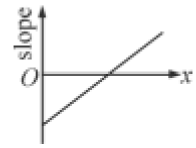
(a) $v_1 - v_2 = v_0$ (b) $v_1 = v_2$ (c) $v_1 + v_0 = v_2$ (d) None of these

4. A heavy particle is projected with a velocity at an angle with the horizontal into the uniform gravitational field. The slope of the trajectory of the particle varies with its horizontal displacement x as :

(a) 

(b) 

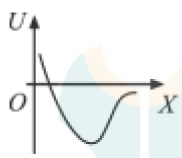
(c) 

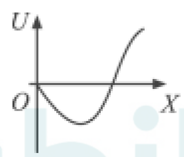
(d) 

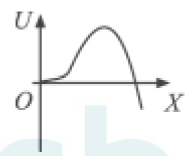
5. A disc is rotating in a room. A boy standing near the rim of the disc of radius R finds the water droplet falling from the ceiling is always falling on his head. As one drop hits his head, other one starts from the ceiling. If height of the roof above his head is H, then angular velocity of the disc is :

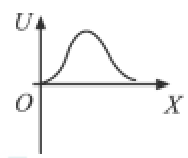
(a) $\pi\sqrt{\frac{2gR}{H^2}}$ (b) $\pi\sqrt{\frac{2gH}{R^2}}$ (c) $\pi\sqrt{\frac{2g}{H}}$ (d) $2\pi\sqrt{\frac{g}{H}}$

6. A particle free to move along x - axis is acted upon by a force $F = -ax + bx^2$ where a and b are positive constants. For $x \geq 0$, the correct variation of potential energy function U(x) is best represented by (take U(0) = 0).

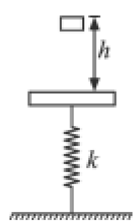
(a) 

(b) 

(c) 

(d) 

7. A vertical spring is fixed to one of its end and a massless plank fitted to the other end. A block is released from a height h as shown. Spring is in relaxed position. Then choose the correct statement.



(a) The maximum compression of the spring does not depend on h

(b) The maximum kinetic energy of the block does not depend on h

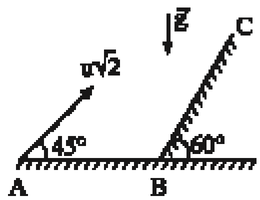
(c) The compression of the spring at maximum KE of the block does not depend on h

(d) The maximum compression of the spring does not depend on k

8. A person slowly pulls a bucket of water from a well of depth h. If the mass of uniform rope is m and that of the bucket full of water is M, then work done by the person is:

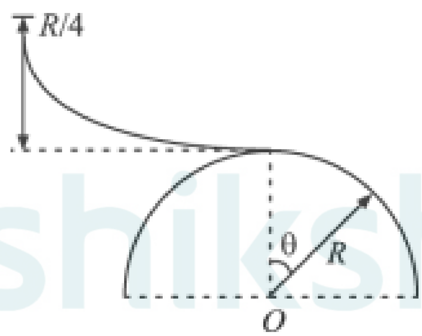
(a) $\left(M+\dfrac{m}{2}\right)gh$ (b) $\dfrac{1}{2}(M+m)gh$ (c) $(M+m)gh$ (d) $\left(\dfrac{M}{2}+m\right)gh$

9. A particle is projected from point 'A' with velocity at an angle of 45° with the horizontal as shown in the figure. It strikes the inclined plane BC at right angle. The velocity of the particle just before collision with the incline is :



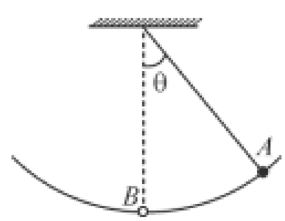
(a) $\dfrac{\sqrt{3}u}{2}$ (b) $\dfrac{u}{2}$ (c) $\dfrac{2u}{\sqrt{3}}$ (d) u

10. A skier plans to ski a smooth fixed hemisphere of radius R. He starts form rest on a curved smooth surface of height $\left(\dfrac{R}{4}\right)$. The angle θ at which he leaves the hemisphere is :



(a) $\cos^{-1}\left(\dfrac{2}{3}\right)$ (b) $\cos^{-1}\dfrac{5}{\sqrt{3}}$ (c) $\cos^{-1}\left(\dfrac{5}{6}\right)$ (d) $\cos^{-1}\left[\dfrac{5}{2\sqrt{3}}\right]$

11. A ball attached to one end of a string swings in a vertical plane such that its magnitude of acceleration at point A (extreme position) is equal to its magnitude of acceleration at point B (mean position). The angle θ is



(a) $\cos^{-1}\left(\dfrac{2}{5}\right)$ (b) $\cos^{-1}\left(\dfrac{4}{5}\right)$ (c) $\cos^{-1}\left(\dfrac{3}{5}\right)$ (d) None of these

12. Which one of the following is true in the case of inelastic collisions ?

Total Energy	Kinetic Energy	Momentum
(a) Conserved	Conserved	Conserved
(b) Conserved	Not Conserved	Conserved
(c) Conserved	Conserved	Not Conserved
(d) Not Conserved	Not Conserved	Conserved

13. In system of particles, internal forces can change:

(a) The linear momentum but not the kinetic energy

(b) The kinetic energy but not the linear momentum

(c) Linear momentum as well as kinetic energy

(d) Neither the linear momentum nor the kinetic energy

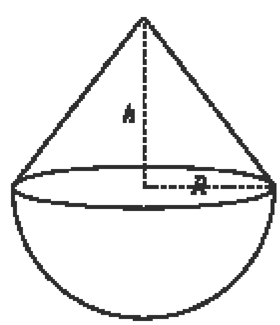
14. From a point on smooth floor of a room, a ball is shot to hit a wall. The ball then returns back to the point of projection. If the time taken by the ball in returning is twice the time taken in reaching the wall, find the coefficient of restitution.

(a) $e = \frac{1}{2}$ (b) $e = \frac{1}{3}$ (c) $e = \frac{1}{4}$ (d) $e = 0.2$

15. A particle of mass m_1 makes on elastic, one dimensional collision with a stationary particle of mass m_2 . What fraction of the kinetic energy of m_1 is carried away by m_2 ?

(a) $\frac{m_1}{m_2}$ (b) $\frac{m_2}{m_1}$ (c) $\frac{2m_1m_2}{(m_1+m_2)^2}$ (d) $\frac{4m_1m_2}{(m_1+m_2)^2}$

16. A uniform solid right circular cone of base radius R is joined to a uniform solid hemisphere of radius R and of the same density, so as to have a common face. The centre of mass of the composite solid lies on the common face. The height of the cone is :



(a) 1.5 R (b) $\sqrt{3}$ R (c) 3R (d) $2\sqrt{3}$ R

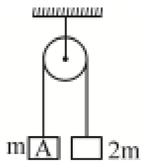
17. A lift is descending with a constant velocity 'v'. A man in the lift drops a coin. The coin experiences an acceleration towards the floor equal to :

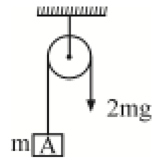
(a) $g + v$ (b) $g - v$ (c) g (d) zero

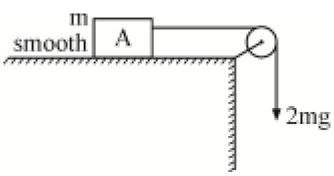
18. A ball is projected horizontal with a velocity of $5ms^{-1}$ from the top of a building 20m. Displacement of ball when it hits the ground is : ($g = 10m/s^2$)

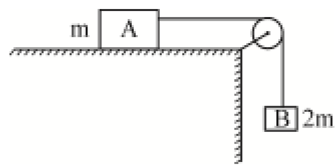
(a) 10 m (b) 20 m (c) $10\sqrt{3}m$ (d) $10\sqrt{5}m$

19. In which of the following cases the magnitude of acceleration of the block A will be maximum: (Neglect friction, mass of pulley and string)

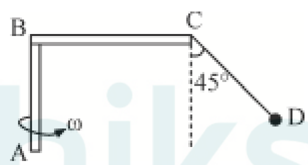
(a) 

(b) 

(c) 

(d) 

20. A ball of mass 1 kg is connected with a L-shaped rod rotating with constant angular velocity $\omega = \sqrt{10}$ rad/s through a light string CD. The length of part BC of the rod is 0.5 m. The length of string CD is : ($g = 10\ m/s^2$)

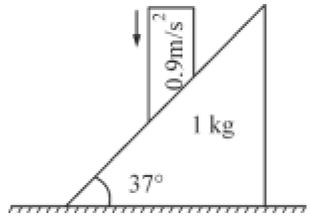


(a) 0.5 m (b) $\frac{1}{2\sqrt{2}}m$ (c) $\frac{1}{\sqrt{2}}m$ (d) $\sqrt{2}m$

SECTION – 2

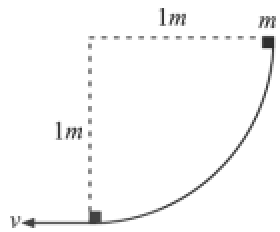
Section 2 contains 5 Numerical Value Type Questions. The answer to each of the question is an integer ranging from 0 to 9.

1. In the system shown in figure, all surfaces are smooth. Rod is moved by external agent with acceleration $0.9\ ms^{-2}$ vertically downwards. Force exerted on the rod by the wedge will be ________ Newtons.



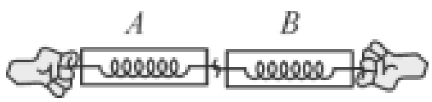
2. A grass hopper can jump maximum distance 1.6 m. It spends negligible time on ground. How far can it go in $2\sqrt{2}s$ (in meters)?

3. A block of mass 1 kg slides down a curved track which forms one quadrant of a circle of radius 1 m as shown in figure. The speed of block at the bottom of the track is v = 2 ms^{-1}. The work done by the force of gravity is 10 x Joules. Find x? (Take g = 10 m/s^2)



4. A particle moves with a velocity $v = (5\hat{i} - 3\hat{j} + 6\hat{k})ms^{-1}$ under the influence of a constant force $F = (\hat{i} - \hat{j} - \hat{k})N$. The instantaneous power applied to the particle is ____________ Watts.

5. Consider two spring balances hooked as shown in the figure. We pull them in opposite directions with equal force. If the reading shown by A is 5 N, reading shown by B will be ________ N.



PART - II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The hybrid state of the Xe and shape of the XeF_4 molecule are :

   (a) sp^3 and tetrahedral (b) sp^3d^2 and tetrahedral

   (c) sp^3 d^2 and square planar (d) sp^3 d and see saw

2. Which of the following relation is correct for an ideal gas under adiabatic reversible conditions ?

   I. PV^{γ} = constant II. $TV^{\gamma-1}$ = constant III. S = constant

   (a) I, II only (b) I, II & III (c) I, III only (d) III only

3. Which of the following is true when a reaction reaches its equilibrium state ?

   I. Entropy of the universe reach its maximum value w.r.t. that reaction

   II. Gibb's free energy of the reaction reach its minimum value at the equilibrium

   III. Rate of forward reaction become equal to rate of backward reaction at the equilibrium

   (a) I, II, III (b) III only (c) II & III only (d) I & II only

4. In which of following conditions a real gas would behave ideally?

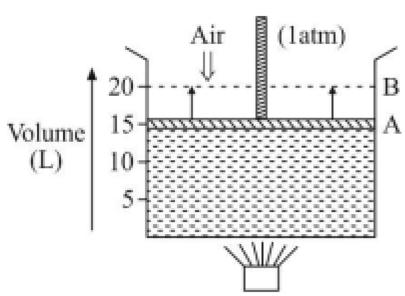
   (a) Low pressure and low temperature

   (b) At value of temperature equal to its Boyle's temperature

   (c) Between its critical temperature and its Boyle's temperature

   (d) At certain high pressure at which compressibility factor Z 1.00

5. An ideal gas absorbs Q amount of heat from the surroundings at constant temperature and uses it completely to undergo a change in its state from A to B against a constant external pressure of 1 atm. The value of Q is (as per the figure given). [Given that 1 atm L 0.1 kJ]



   (a) 0.5kJ (b) 0.5kJ (c) – 1kJ (d) 1kJ

6. For which of following reversible reaction the degree of dissociation or association depends upon the value of temperature but NOT on value of pressure ?

(a) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (b) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(c) $3C_2H_2(g) \rightleftharpoons C_6H_6(g)$ (d) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

7. Which of following is a extensive state variable ?

(a) W (work done)

(b) G (Gibb's free energy)

(c) C_m (molar heat capacity)

(d) $\Delta_f H^o$ (standard molar enthalpy of formation)

8. Which of the following is a set of paramagnetic species ?

(a) $O_2(g), B_2(g)$ (b) $O_2(g), C_2(g)$ (c) $B_2(g), C_2(g)$ (d) H_2, O_2^+

9. Which of the following pair of ideal gases will diffuse at same rate under identical condition of P and T ? (Molar mass of C = 12 g, H = 1 g, O = 16 g, N = 14g, He = 4 g)

(a) $C_2H_4(g)$ and $C_3H_6(g)$ (b) $CO_2(g)$ and $O_3(g)$

(c) He(g) and $H_2(g)$ (d) $N_2H_4(g)$ and $O_2(g)$

10. Given that bond energies of $N \equiv N, H-H$ and N – H are 900 kJ/mole, 350 kJ / mole and 450 kJ/mole respectively. Value of $\Delta_f H^o_{NH_3}$ (g) would be :

(a) -150 kJ / mole (b) -300 kJ / mole (c) -425 kJ / mole (d) -375 kJ / mole

11. Combustion of 1 mole of $C_6H_6(\ell)$ inside a closed container of constant volume liberates 900 kJ of heat energy. What would be the value of heat liberated per mole of $C_6H_6(\ell)$ if the reaction occurs at 300 K in an open vessel?

(a) 903.75 kJ / mole (b) -1803.75 kJ / mole (c) -1796.25 kJ / mole (d) +896.25 kJ / mole

12. To the system $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ in equilibrium, some N_2 gas was added at constant volume. Then :

(a) K_p will remain constant and K_c will change (b) K_c will remain constant and K_p will change

(c) Both K_p and K_c remain constant (d) Both K_p and K_c will change

13. What is the expression for equilibrium constant KC for the following reaction? $2A(s) + 5B(g) \rightleftharpoons 2C(s)$

(a) $K_C = [B]^5$ (b) $K_C = \dfrac{2[C]}{2[A]\times 5[B]}$ (c) $K_C = \dfrac{[C]^2}{[A]^2\times[B]^5}$ (d) $K_C = \dfrac{1}{[B]^5}$

14. Dipole molecule of H – F molecule is 2.4 debye and its bond length is 1.0 Å. If 1 debye is equal to 1×10^{-18} esu × cm, find percentage ionic character in H –F. [charge on an electron $= 4.8 \times 10^{-10}$ esu]

(a) 57% (b) 50% (c) 49% (d) 43%

15. An amount of solid NH_4 HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. Ammonium hydrogen sulphide decomposes to yield NH_3 and H_2S gases in the flask. When the decomposition reaction reaches equilibrium, the total pressure in the flask rises to 0.84 atm. The equilibrium constant (K_p) for NH_4HS decomposition at this temperature is:

$[NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)]$

(a) 0.029 (b) 0.091 (c) 0.40 (d) 0.114

16. Using the given data find value of resonance energy of benzene C_6H_6.

$\Delta_{hydg}H^o$ of cyclohexene = -119 kJ / mole

$\Delta_{hydg}H^o$ of benzene = - 206.5 kJ / mole

(a) -150.5 kJ/mole (b) -325.5 kJ/mole (c) -87.5 kJ/mole (d) -131.5 kJ/mole

17. Select the correct match. (in reference to an ideal gas)

(a) Isothermal reversible process $\Delta S_{sys} = nRT\log\left(\dfrac{P_f}{P_i}\right)$

(b) Isobaric reversible process $\Delta S_{sys} = nC_v\log\left(\dfrac{T_f}{T_i}\right)$

(c) Isothermal irreversible process $\Delta S_{sys} = nR\log\left(\dfrac{V_f}{V_i}\right)$

(d) Adiabatic irreversible process $\Delta S_{sys} = nC_p\log\left(\dfrac{T_f}{T_i}\right)$

18. Select correct alternative in reference to following statements:

I. During formation of a H-bond, the intramolecular H-bonding is less favourable than intermolecular H-bonding

II. According to molecular orbital theory the potential energy of $\sigma 2p_z$ M.O. is always less than that of the $\pi 2p_x$ M.O. on taking Z as the bonding axis

(a) I and II both (b) I only

(c) II only (d) Both I and II are incorrect

19. K_1 and K_2 are equilibrium constants for reaction (1) and (2)

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ ....(1)

$NO(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ ....(2)

Then :

(a) $K_1\left(\frac{1}{K_2}\right)^2$ (b) $K_1 = K_{2^2}$ (c) $K_1 = \frac{1}{K_2}$ (d) $K_1 = 2K_2$

20. A 4 L gas mixture of ethane (C_2H_6) and propane (C_3H_8) on complete combustion at 25°C produced 10 L of CO_2. What is the composition of gas mixture i.e. volume of C_2H_6 and C_3H_6 respectively?

(a) 2L, 2L (b) 1.5 L, 2.5 L (c) 2.5 L, 1.5 L (d) None of these

SECTION – 2

Section 2 contains 5 Numerical Value Type Questions. The answer to each of the question is an integer ranging from 0 to 9.

1. At critical condition of temperature and pressure for a real gas, what would be the value of 16 Z ? Here, Z is its compressibility factor.

2. In the following list of compounds, the number of species whose central atoms use d-orbitals in hybridisation and also have zero dipole moment:

XeF_4 XeF_6 XeF_2 SF_6 PC_5 PF_2Cl_3

3. Given that $\Delta_C H^o$ (combustion enthalpy) for Diamond is -834.8 kJ mole and $\Delta_C H^o$ for graphite is -832.8 kJ/mole, the value of $\Delta_f H^o_{(Diamond)}$ i.e. [C(graphite) → C(diamond)] [in kJ/mole] is:

4. Certain fixed quantity of an ideal gas expands by adiabatic manner such that w_{gas} is equal to -25 kJ and ΔH_{gas} for the expansion is -75kJ. The ratio of $\frac{C_p}{C_v}$ for this gas will be equal to:

5. For a reversible reaction $AB(g) + CE(g) \rightleftharpoons AD(g) + BE(g)$ give that :

$\Delta_f H_{(AB)} = -40 kJ/mole$ $\Delta_f H_{(CE)} = -20 kJ/mole$

$\Delta_f H_{(AD)} = -30\text{kJ/mole}$ $\Delta_f H_{(BE)} = -10\text{kJ/mole}$

Entropy $S_{AB} = 5\text{J/K}$ $S_{CE} = 8\text{J/K}$ $S_{AD} = 15\,\text{J/K}$ $S_{BE} = 18\,\text{J/K}$

To equilibrium temperature for the reaction is 'T' K. What is the value of $\frac{T}{1000}$?

PART - III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Letter of the word INDIANOIL are arranged in all possible ways. The number of permutations in which A, I, O occur only at odd places, is :

   (a) 720 (b) 360 (c) 240 (d) 120

2. How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S are adjacent ?

   (a) $(8)\left({}^6C_4\right)\left({}^7C_4\right)$ (b) $(8)(7)\left({}^7C_4\right)$ (c) $(6)(8)\left({}^7C_4\right)$ (d) $(7)\left({}^6C_4\right)\left({}^8C_4\right)$

3. The letters of the word COCHIN are permutated and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before COCHIN is :

   (a) 360 (b) 192 (c) 96 (d) 48

4. A man invites 6 non-vegetarian friends and 5 vegetarian friends for a dinner party. He arranges 6 nonvegetarian friends on one round table and 5 vegetarian friends along another round table. The number of ways this can be done is:

   (a) 11! (b) 9! (c) 2880 (d) 8280

5. m men and w women are to be seated in a row so that all women sit together. The number of ways in which they can be seated is :

   (a) (m + 1)! (b) m! w! (c) m! (w-1)! (d) (m + 1)! W!

6. The number of (staircase) paths in the xy-plane from (0, 0) to (7, 5) where each such path is made up of individual steps going one unit upward (U) or one unit to the right (R).

   (a) ${}^{12}C_5$ (b) 12! (c) 5! 7! (d) ${}^{12}P_5$

7. The number of integral solutions of x + y + z = 0 with $x-5, y \geq -5, z \geq -5$ is:

(a) 134 (b) 136 (c) 138 (d) 140

8. There are three piles of identical yellow, black and green balls and each pile contains at least 20 balls. The number of ways of selecting 20 balls if the number of black balls to be selected is twice the number of yellow balls, is :

(a) 6 (b) 7 (c) 8 (d) 9

9. Kunal Gaba has n objects, each of weight w. He weighs them in pairs and finds the sum of the weights of all possible pairs is 120 g. When his friend Rakshit weighs them in triplets, the sum of all possible weights is 240 g. The value of n is :

(a) 7 (b) 6 (c) 5 (d) 10

10. If $|\omega| = 2$, then the set of points $x + iy = \omega - \frac{1}{\omega}$ lie on :

(a) circle (b) ellipse (c) parabola (d) hyoperbola

11. If $a, b, x, y \in R, \omega \neq 1$, is a cube root of unity and $(a + b\omega)^7 = x + y\omega$, then $(b + a\omega)^7$ equals:

(a) $y + x\omega$ (c) $-y - x\omega$ (c) $y + y\omega$ (d) $-x - y\omega$

12. The value of $S = \sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is :

(a) -1 (b) 0 (c) – i (d) i

13. If $\omega = \cos\frac{\pi}{n} + i\sin\frac{\pi}{n}$, then value of $1 + \omega + \omega^2 + \ldots + \omega^{n-1}$ is:

(a) $1 + i\cot\left(\frac{\pi}{2n}\right)$ (b) $1 + i\tan\left(\frac{\pi}{2n}\right)$ (c) $1 + i$ (d) None of these

14. If $z \in C - \{0, -2\}$ is such that $\log_{(1/7)}|z - 2| > \log_{(1/7)}|z|$, then :

(a) $\text{Re}(z) > 1$ (b) $\text{Re}(z) < 1$ (c) $\text{Im}(z) > 1$ (d) $\text{Im}(z) < 1$

15. If $\omega \neq 1$ is a cube root of unity, then $z\sum_{k=1}^{60}\omega^k - \prod_{k=1}^{30}\omega^k$ is equal to :

(a) 0 (b) ω (c) ω^2 (d) -1

16. Let a_1, a_2, ….. be in H.P. with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is :

(a) 22 (b) 23 (c) 24 (d) 25

17. Sum of the series $\sum_{r=1}^{n} r\log\frac{r+1}{r}$ is :

(a) $\log \dfrac{(n+1)^n}{n!}$ (b) $\log \dfrac{(n+1)}{n!}$ (c) $n!\log(n+1)$ (d) None of these

18. Suppose a, b, c are distinct positive real numbers such that a, b, c 2 3 are in A.P. and a, b, c are in G.P. The common ratio of G.P. is :

(a) 2 (b) $\dfrac{1}{2}$ (c) $\dfrac{1}{3}$ (d) 3

19. The sum to n terms of the series $\dfrac{1}{1^3}+\dfrac{1+2}{1+2^3}+\dfrac{1+2+3}{1^3+2^3+3^3}+....$ is:

(a) $\dfrac{n}{n+1}$ (b) $\dfrac{n}{2(n+1)}$ (c) $\dfrac{2n}{n+1}$ (d) $\dfrac{2}{n(n+1)}$

20. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If a < b < c and $a+b+c=\dfrac{3}{2}$, then the value of a is :

(a) $\dfrac{1}{2\sqrt{2}}$ (b) $\dfrac{1}{2\sqrt{3}}$ (c) $\dfrac{1}{2}-\dfrac{1}{\sqrt{2}}$ (d) None of these

SECTION – 2

Section 2 contains 5 Numerical Value Type Questions. The answer to each of the question is an integer ranging from 0 to 9.

1. The value of $\sum_{n=1}^{49} \underline{|n}(n^2+n+1)$ is equal to $(10\alpha)\underline{|10}\alpha - ß$. The value of $(\alpha + ß)$ is ____________.

2. If $\dfrac{48}{(2)(3)}+\dfrac{47}{(3)(4)}+\dfrac{46}{(4)(5)}+...+\dfrac{2}{(48)(49)}+\dfrac{1}{(49)+(50)}=\dfrac{51}{2}+k\left(1+\dfrac{1}{2}+\dfrac{1}{3}+...+\dfrac{1}{50}\right)$. Then $|k|$ is equal to ___________.

3. In a geometric progression the ratio of the sum of the first 5 terms to the sum of their reciprocals is 49 and sum of the first and the third term is 35. The fifth term of the G.P. is p, the value of 4p is _______.

4. 5-digit numbers are formed using 2, 3, 5, 7, 9 without repeating the digits. If p is the number of such numbers that exceed 20000 and q be the number of those that lie between 30000 and 90000 then $\dfrac{3p}{q}$ is __________.

5. Rakshit is allowed to select 1 or more books out of (2n + 1) distinct books. If the number of ways in which he may not select all of them is 126, then value of n is _________.